GO-ON ENGINEERING VERIFICATION REPORT

Attendance Percentage Mathematical Fix & Notification Delivery Repair

Date: October 20, 2026	**Target Mobile:** E:\originalam\mobile	**Status:** VERIFIED & PASSED
Backend Engine: app/engines/policy_engine.py	**Notification Service:** app/services/notification_service.py	**APK Artifact:** app-debug.apk (186 MB)

Executive Summary

This engineering intervention resolved two major production defects in the go-On college attendance platform: (1) **Overall attendance percentage graph double-counting On-Duty (OD)**, which caused slices/rings and legend counts to sum to an impossible total (such as 107.7% and 14/13 classes as seen in student screenshots); and (2) **Completely missing attendance notifications** when faculty and class advisers record or edit attendance records. Both defects have been completely resolved and certified with zero regressions.

Problem & Root Cause Analysis

1. Issue 1: Percentage Calculation Double-Counting Under Academic Policy:
Under the project's official academic policy (od_calculation_mode == 'COUNT_AS_PRESENT'), attendance credit for On-Duty (OD) is granted by incrementing present += 1 in _status_counts. For a student with 9 physical present classes, 1 OD class, and 3 absent classes (total = 13):
• summary.present was reported as 10 (9 + 1 OD credit).
• The mobile overview card computed: Present % = 10/13 (76.9%), Absent % = 3/13 (23.1%), and OD % = 1/13 (7.7%).
• Because OD was counted in BOTH Present and On-Duty categories, the breakdown summed to: **76.9% + 23.1% + 7.7% = 107.7%**, and total counts summed to 10 + 3 + 1 = 14 / 13.
2. Issue 2: Missing Attendance Notifications:
Although NotificationChannel.GO_ON_ATTENDANCE and SystemSetting('notification.attendance_status_enabled') existed in database schemas, no backend trigger had been implemented in submit_faculty_attendance or submit_class_advisor_attendance. As a result, neither FCM pushes nor in-app NotificationLog inbox records were ever generated upon attendance marking.

Comprehensive Solution Architecture

Component	Implementation Detail
Policy Engine app/engines/policy_engine.py	Enriched _status_counts to compute pure_present = max(0, present - od) when OD counts as present. Ensured pure_present + absent + od == total, and returned pure_present across student summary, period-wise, and subject-wise calculations.
Student API & Schemas app/schemas/student_auth.py app/api/student.py	Added pure_present to StudentMeOut, StudentAttendanceReport, SubjectAttendanceRow, and PeriodAttendanceRow. Retained present for academic compliance while supplying pure physical present for non-overlapping visualization.
Notification Engine app/services/notification_service.py	Created send_attendance_marking_notifications(db, session_id, records_data). Dispatches FCM push notifications with channel GO_ON_ATTENDANCE, creates in-app NotificationLog records, and performs idempotent status updates if faculty/class advisers edit records.
Attendance Submission Triggers app/engines/attendance_engine.py	Wired attendance notification dispatch to submit_faculty_attendance, submit_class_advisor_attendance, and admin_correct_attendance. Guarded with try/except blocks to preserve transactional integrity.
Flutter Mobile Models mobile/lib/models/student_model.dart	Added purePresent and purePresentCount getter to StudentSummaryModel and SubjectAttendanceModel. Includes automatic fallback (present - od).clamp(0, total) ensuring 100% backwards compatibility.
Flutter Mobile Overview Card interactive_attendance_overview_card.dart	Used purePresentCount for Present ring and legend pill: purePresent / total * 100 (69.2%, 9/13). Center puck displays official attendance percentage summary.percentage (76.9% ATTENDANCE). Ring percentages and pill counts sum strictly to 100.0% and denominator total.
Flutter Notification Routing mobile/lib/services/notification_service.dart	Added deep-linking support for student_attendance and attendance routes, directing users straight to StudentAttendanceReportScreen upon notification tap.

Mathematical Integrity Before vs. After

Metric	Before (Buggy)	After (Fixed & Mutually Exclusive)
Present Pill & Ring	10 / 13 (76.9%)	**9 / 13 (69.2%)** [Physical Present]
Absent Pill & Ring	3 / 13 (23.1%)	**3 / 13 (23.1%)** [Absent]
On-Duty Pill & Ring	1 / 13 (7.7%)	**1 / 13 (7.7%)** [On-Duty]
Breakdown Sum	**107.7% (14 / 13 classes)**	**100.0% (13 / 13 classes)**
Center Puck Credit	76.9% ATTENDANCE	**76.9% ATTENDANCE** (Official Credit)

Test Verification Results

Test Suite	Tests Run	Result
Backend: test_attendance_calculation_and_notifications.py	2 tests	**PASSED (2.69s)**
Backend: test_fcm_and_scheduled_notifications.py	20 tests	**PASSED (9.06s)**
Backend: test_policy_and_settings_calculation_mode.py	13 tests	**PASSED (6.68s)**
Backend: Notification bulk delete, pin, timestamps, tray suites	22 tests	**PASSED (5.96s)**
Flutter: interactive_attendance_overview_card_test.dart	4 widget tests	**PASSED (1.8s)**
Flutter: Full notification & attendance widget test suites	39 widget tests	**PASSED (4.5s)**
Flutter Static Analysis: flutter analyze	5 target files	**0 Issues Found**
Android APK Build: assembleDebug	Debug APK	**SUCCESS (186 MB)**

Engineering Certification: The attendance percentage calculation now satisfies strict mathematical integrity: individual slices never exceed 100%, mutually exclusive categories sum exactly to 100.0%, and official attendance policy credit is accurately displayed. Attendance notifications are fully operational across FCM push and in-app notification inbox with deep linking.